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# Computational Discovery of DNA Aptamer-Graphene Sensors

august 2026

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Master's Dissertation in Bioinformatics

Dissertation supervised by
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august 2026

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University of Minho, Braga, august 2026

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# Abstract

Rapid and reliable detection of microbial DNA is central to clinical diagnostics, antimicrobial-resistance surveillance and environmental monitoring, yet the dominant amplification-based methods require laboratory infrastructure that limits their use at the point of care. Graphene field-effect transistor (GFET) biosensors, functionalised with single-stranded DNA recognition elements, offer a compelling label-free alternative, converting the hybridisation of a target sequence at the graphene surface into a measurable electrical signal. The bottleneck of this approach is the recognition sequence: identifying, for each pathogen, probe candidates that are conserved across strains, hybridise favourably, and are free of stable self-structure, without resorting to costly and low-throughput experimental screening of hundreds of candidates.

This dissertation addresses that bottleneck with a reproducible computational pipeline for the *discovery* of DNA probe candidates for GFET biosensors, developed for a panel of clinically relevant bacterial marker genes. From sequences retrieved from the NCBI databases, the pipeline performs automatic length clustering, multiple sequence alignment with MAFFT, and per-column conservation scoring based on the Percentage of Pairwise Identity. Sliding-window candidates are screened with nearest-neighbour thermodynamics (melting temperature, GC content, hairpin and homodimer free energies) and evaluated for secondary structure, condensed into a continuous "No-fold" score. All thresholds are resolved through an organism-aware, layered scheme (gene, species and organism type), so that pathogens with very different genome composition are judged fairly. The highest-quality candidates are prioritised and validated three-dimensionally, and a curated, feature-rich dataset of the best candidates per gene is produced.

Applied to the six-gene panel, the pipeline reduced 8455 candidate windows to 2943 that passed thermodynamic and structural screening, from which the best probes per gene were validated in 3D. Alignment-free diversity and rarefaction analyses characterised each target and justified the sampling depth. The result is a prioritised, biophysically sound set of probe candidates and a reproducible framework that provides the sequence-level foundation for the rational design of aptamer–graphene GFET biosensors, and that generalises to new genes and organisms with minimal configuration.

**Keywords** DNA aptamer, DNA probe, graphene biosensor, graphene field-effect transistor (GFET), label-free detection, microbial DNA detection, point-of-care diagnostics, probe design, sequence conservation, multiple sequence alignment, melting temperature, secondary structure, No-fold score, 3D structure prediction

# Resumo

A deteção rápida e fiável de DNA microbiano é central para o diagnóstico clínico, a vigilância da resistência antimicrobiana e a monitorização ambiental, mas os métodos dominantes, baseados em amplificação, exigem infraestrutura laboratorial que limita o seu uso no local de prestação de cuidados. Os biossensores baseados em transístores de efeito de campo de grafeno (GFET), funcionalizados com elementos de reconhecimento de DNA de cadeia simples, oferecem uma alternativa sem marcação apelativa, convertendo a hibridação de uma sequência-alvo à superfície do grafeno num sinal elétrico mensurável. O estrangulamento desta abordagem é a sequência de reconhecimento: identificar, para cada agente patogénico, sondas candidatas que sejam conservadas entre estirpes, hibridem favoravelmente e estejam livres de estrutura secundária estável, sem recorrer à triagem experimental dispendiosa e de baixo débito de centenas de candidatas.

Esta dissertação aborda esse estrangulamento com um pipeline computacional reprodutível para a *descoberta* de sondas de DNA candidatas para biossensores GFET, desenvolvido para um painel de genes marcadores bacterianos clinicamente relevantes. A partir de sequências obtidas nas bases de dados do NCBI, o pipeline realiza a seleção automática do agrupamento de comprimento, o alinhamento múltiplo com o MAFFT e a pontuação de conservação por coluna baseada na Percentagem de Identidade Par-a-Par. As candidatas, geradas por janela deslizante, são filtradas com termodinâmica de vizinhos mais próximos (temperatura de fusão, conteúdo em GC, energias de hairpin e homodímero) e avaliadas quanto à estrutura secundária, condensada num score contínuo "No-fold". Todos os limiares são resolvidos através de um esquema em camadas e sensível ao organismo (gene, espécie e tipo de organismo), para que agentes patogénicos com composição genómica muito distinta sejam avaliados de forma justa. As candidatas de maior qualidade são priorizadas e validadas tridimensionalmente, e é produzido um conjunto de dados curado e rico em atributos com as melhores candidatas por gene.

Aplicado ao painel de seis genes, o pipeline reduziu 8455 janelas candidatas a 2943 que passaram a triagem termodinâmica e estrutural, a partir das quais as melhores sondas por gene foram validadas em 3D. Análises de diversidade sem alinhamento e de rarefação caracterizaram cada alvo e justificaram a profundidade de amostragem. O resultado é um conjunto priorizado e biofisicamente sólido de sondas candidatas e uma estrutura reprodutível que constitui o alicerce, ao nível da sequência, para o desenho racional de biossensores GFET de aptâmero–grafeno, generalizável a novos genes e organismos com configuração mínima.

**Palavras-chave** aptâmero de DNA, sonda de DNA, biossensor de grafeno, transístor de efeito de campo de grafeno (GFET), deteção sem marcação, deteção de DNA microbiano, diagnóstico no local de prestação de cuidados, desenho de sondas, conservação de sequência, alinhamento múltiplo de sequências, temperatura de fusão, estrutura secundária, score No-fold, previsão de estrutura 3D

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# Acronyms

$T_m$ melting temperature.

**AMR** Antimicrobial Resistance.

**FRET** Förster Resonance Energy Transfer.

**GC** Guanine-Cytosine.

**GFET** Graphene Field-Effect Transistor.

**GO** Graphene Oxide.

**GPU** Graphics Processing Unit.

**IUPAC** International Union of Pure and Applied Chemistry.

**MFE** Minimum Free Energy.

**MSA** Multiple Sequence Alignment.

**NCBI** National Center for Biotechnology Information.

**NN** Nearest-Neighbour.

**PAE** Predicted Aligned Error.

**PCR** Polymerase Chain Reaction.

**pLDDT** predicted Local Distance Difference Test.

**PoC** Point-of-Care.

**PPI** Percentage of Pairwise Identity.

**SELEX** Systematic Evolution of Ligands by EXponential enrichment.

**ssDNA** single-stranded DNA.

# Chapter 1

# Introduction

## 1.1 Context

Bacterial infections are among the leading causes of death worldwide, and the spread of **Antimicrobial Resistance (AMR)** is steadily eroding the ability to treat them: an estimated $4.95$ million deaths were associated with bacterial **AMR** in $2019$, of which $1.27$ million were directly attributable to it (Murray et al., 2022). Rapid identification of the infecting pathogen, and increasingly of its resistance determinants, is therefore central to guiding therapy, to infection control and to public-health surveillance; several of the species targeted in this work feature on the WHO Bacterial Priority Pathogens List (World Health Organization, 2024).

The fast identification of microbial DNA underpins the diagnosis of infection and the tracking of environmental contamination. Culture-based methods are slow, and although the **Polymerase Chain Reaction (PCR)** and related amplification techniques are highly sensitive and specific, they require dedicated equipment, controlled laboratory conditions and lengthy sample preparation, which restricts their use outside the laboratory (Hu et al., 2024). This has driven the development of miniaturised solid-state biosensors, which convert a molecular-recognition event directly into an electrical or electrochemical signal and thereby enable rapid, portable and **Point-of-Care (PoC)** testing (Kozel and Burnham-Marusich, 2017; Krishnan et al., 2023).

Coupling DNA recognition elements to a **Graphene Field-Effect Transistor (GFET)** has been proposed as a particularly promising route to label-free nucleic-acid detection (Green and Norton, 2015). Graphene, a two-dimensional sheet of carbon isolated in 2004 (Novoselov et al., 2004), has its entire conduction channel exposed at its surface and transduces local charge variations with such sensitivity that even a single adsorbed molecule can be detected (Schedin et al., 2007); when it is functionalised with a **single-stranded DNA (ssDNA)** probe, the hybridisation of a complementary microbial target modulates the transistor current in a label-free manner. In practice, the probe is anchored to the graphene surface through a linker, such as a pyrene derivative that binds non-covalently by $\pi$–$\pi$ stacking, and target hybridisation shifts the transistor's charge-neutrality (Dirac) point; electrolyte-gated graphene devices of this kind have achieved label-free detection of DNA hybridisation down to the attomolar range (Campos et al., 2019). The performance of such devices, and the influence of the sensing interface and ionic conditions on it, is reviewed in detail in Chapter 2.

The specificity and sensitivity of such a device ultimately depend on the immobilised recognition sequence, which must meet several partly competing requirements: it must target a genomic region that is conserved across the clinical strains of the pathogen, so that strain variation does not cause false negatives; it must hybridise strongly and specifically at the assay temperature, which constrains its length, melting temperature and **GC** content; and it must avoid folding into stable hairpins or self-dimers that would compete with hybridisation. These requirements are biophysical and quantifiable, which makes them amenable to systematic computational evaluation.

Building on these experimental developments, a central practical obstacle remains: the high cost and low throughput of experimentally testing hundreds of candidate probes in order to discriminate multiple pathogenic species. This motivates a *complementary computational screening strategy* that can prioritise a small number of promising candidates before any experimental effort is spent.

## 1.2 Problem statement

The scientific problem addressed in this dissertation is how to computationally *discover and prioritise* DNA probe candidates that recognise specific microbial DNA sequences and are suitable for transduction on a **GFET**. This requires analysing the sequence and structural features — genomic conservation, hybridisation thermodynamics and self-structure — that determine whether a probe will bind its target strongly and specifically across clinical strains, and doing so in a reproducible, automated way that scales across many genes and organisms of very different genome composition (Green and Norton, 2015; Campos et al., 2019).

## 1.3 Objectives

The main objective is to build a reproducible computational pipeline that turns a target gene into a small set of ranked, biophysically validated **ssDNA** probe candidates for **GFET** biosensing. This decomposes into the following specific objectives:

- select a panel of clinically relevant microbial targets and marker genes, and automate the retrieval and curation of their sequences from the **NCBI** databases;
- identify conserved regions through multiple sequence alignment and per-column conservation scoring, so that the probes detect their target across clinical strains;
- screen candidate probes with well-founded biophysical criteria (melting temperature, GC content, hairpin, homodimer and secondary-structure stability) and combine them into transparent quality scores;
- make the design criteria *organism-aware*, so that targets with very different genome composition (AT-rich or GC-rich) are evaluated fairly;
- prioritise and validate the most promising candidates by three-dimensional structure prediction; and
- produce curated, feature-rich datasets of the prioritised probe candidates.

## 1.4 Target panel

Six marker genes from six clinically important bacterial species were used as the working panel throughout this work (Table 1). Several are established targets for the molecular detection of their species: the thermonuclease gene *nuc* for *Staphylococcus aureus* (Brakstad et al., 1992), the autolysin gene *lytA* for *Streptococcus pneumoniae* (Carvalho et al., 2007), and the outer-membrane lipoprotein gene *oprL* for *Pseudomonas aeruginosa* (De Vos et al., 1997). The panel spans a wide range of genome composition — from AT-rich (*Staphylococcus aureus*, *Haemophilus influenzae*) to GC-rich (*Pseudomonas aeruginosa*) — which makes it a demanding test of a general-purpose design method.

| Gene | Organism | Group | Function |
|---|---|---|---|
| *nuc* | *Staphylococcus aureus* | A | thermonuclease |
| *rmpM* | *Neisseria meningitidis* | A | class-4 outer-membrane protein |
| *lytA* | *Streptococcus pneumoniae* | B | autolysin |
| *oprL* | *Pseudomonas aeruginosa* | B | peptidoglycan-associated lipoprotein |
| *algD* | *Pseudomonas aeruginosa* | B | GDP-mannose 6-dehydrogenase |
| *frdB* | *Haemophilus influenzae* | B | fumarate reductase Fe–S subunit |

Table 1: Target genes and their host organisms used as the working panel.

## 1.5 Structure of the dissertation

The remainder of this dissertation is organised as follows. Chapter 2 reviews the state of the art in DNA aptamers and probes, in graphene **GFET** biosensors for nucleic-acid detection, and in the computational methods used to design and validate probe sequences. Chapter 3 describes the methods, detailing every stage of the pipeline and the rationale for each design decision and threshold. Chapter 4 presents and discusses the results for the target panel — the screening funnel, the three-dimensional validation, the diversity and rarefaction analyses, and the comparison with a reference probe panel. Chapter 5 draws the main conclusions and outlines future work.

# Chapter 2

# State of the Art

This chapter reviews the background needed to frame the work. It covers DNA aptamers and probes as recognition elements, graphene **GFET** biosensors for nucleic-acid detection, the physics of the DNA–graphene interface, and the computational methods used to design and validate probe sequences.

## 2.1 DNA aptamers and probes as recognition elements

DNA aptamers are short single-stranded nucleic acids, selected *in vitro* by **Systematic Evolution of Ligands by EXponential enrichment (SELEX)** (Tuerk and Gold, 1990; Ellington and Szostak, 1990), that fold into defined structures capable of high-affinity, high-specificity recognition of a target (Stephens et al., 2025; Sun et al., 2022). Cell-**SELEX** and related variants allow aptamers to be raised against whole microbial cells and complex targets, yielding recognition elements for bacterial surface antigens, toxins and viral components (Kolm et al., 2020; Hu et al., 2024). Because they are chemically synthesised, aptamers and DNA probes are inexpensive, reproducible and easily modified, which makes them well suited to solid-state biosensors and to *in silico* optimisation (Sun et al., 2022). For the detection of a known microbial DNA sequence, a complementary hybridisation probe is the natural recognition element: it targets a defined genomic region and is captured by Watson–Crick pairing, which is the mechanism exploited throughout this work.

## 2.2 Graphene GFET biosensors for DNA detection

Graphene combines very high carrier mobility with a fully surface-exposed conduction channel, so that charges binding at its surface strongly modulate its conductance (Novoselov et al., 2004); the adsorption of even a single molecule can be resolved electrically (Schedin et al., 2007), making it an outstanding transducer for label-free sensing (Krishnan et al., 2023; Green and Norton, 2015), including for nucleic acids at very low concentrations (Hwang et al., 2020). In a **Graphene Field-Effect Transistor (GFET)**, the graphene channel is gated electrolytically, and the hybridisation of a target strand to an immobilised probe shifts the transfer characteristic through the added backbone charge. Electrolyte-gated **GFET**s functionalised with DNA probes have reached attomolar, label-free detection of hybridisation (Campos et al., 2019), back-gated architectures have enabled scalable device production (Ping et al., 2016), and microfluidic integration has turned such devices into programmable lab-on-chip DNA sensors (Purwidyantri et al., 2022). A key operating constraint is the ionic strength of the medium: high salt concentration produces strong Debye screening that shortens the effective sensing length and attenuates the signal from charges far from the surface (Purwidyantri et al., 2021); transduction schemes based on target-induced conformational change of the probe have been shown to mitigate this limitation (Nakatsuka et al., 2018). These experimental studies define the device context that the computational design in this dissertation aims to support.

## 2.3 The DNA–graphene interface

The behaviour of a **GFET** DNA sensor is governed by how DNA interacts with graphene. Single-stranded DNA adsorbs strongly and flat onto graphene through $\pi$–$\pi$ stacking of its exposed nucleobases, whereas double-stranded DNA binds much more weakly because base pairing sequesters the bases (Green and Norton, 2015; Krishnan et al., 2023). This contrast is the basis of many transduction schemes: for example, graphene-oxide **Förster Resonance Energy Transfer (FRET)** aptasensors, in which a single-stranded aptamer adsorbed on **Graphene Oxide (GO)** is desorbed upon target binding, restoring fluorescence (Lu et al., 2009; Zhu et al., 2015). Adsorption can, however, perturb DNA folding and kinetics, which matters especially for structured recognition elements: a useful probe must retain its ability to hybridise its target while still producing a detectable electronic perturbation at the surface (Green and Norton, 2015).

## 2.4 Computational design and validation of probe sequences

Because experimentally screening large numbers of candidate probes is costly and slow, computational methods that link sequence and structure to binding and transduction are increasingly used to design and prioritise candidates (Sun et al., 2022). Several families of criteria are well established. The stability of the probe–target duplex is quantified by its melting temperature ($T_m$), predicted accurately with the nearest-neighbour thermodynamic model (SantaLucia Jr. and Hicks, 2004), together with GC content and probe length (Wetmur, 1991); tools such as `primer3` implement these calculations (Untergasser et al., 2012). Self-structure — hairpins, self-dimers and the overall **Minimum Free Energy (MFE)** of the single strand — must be minimised, since a folded probe is unavailable for capture; these energies are obtained from nearest-neighbour folding models (Turner and Mathews, 2010; Zadeh et al., 2011; Timmons, 2020). To detect a pathogen across its strains, conserved genomic regions are identified from a **Multiple Sequence Alignment (MSA)** of many sequences, for which MAFFT is a fast and accurate aligner (Katoh and Standley, 2013); conservation is then scored per column, for example by the **Percentage of Pairwise Identity (PPI)** adopted from the *ViruScope* tool (Lima, 2023).

Beyond sequence, deep-learning methods now predict three-dimensional structure with high accuracy. AlphaFold achieved near-experimental protein predictions (Jumper et al., 2021), and AlphaFold 3 extended this to nucleic-acid-containing complexes with per-residue and pairwise confidence estimates (Abramson et al., 2024); open models such as Boltz-1/2 make these predictions freely runnable on commodity GPUs (Wohlwend et al., 2024). Although calibrated primarily for proteins, the predicted structure and the associated confidence of a short **ssDNA** probe provide a useful, independent check on a computationally selected sequence.

## 2.5 Summary

Graphene **GFET** devices are an excellent, experimentally proven transducer for label-free DNA detection, but their performance hinges on the recognition sequence. The biophysical criteria for a good probe — conservation, favourable hybridisation thermodynamics and absence of stable self-structure — are individually well understood and supported by mature software, yet they are rarely combined into a single, reproducible and organism-aware pipeline that also ranks the candidates and validates them structurally. The methodology presented in the next chapter is designed to address that gap.

# Chapter 3

# Methods

The probe-discovery method was implemented as a single, reproducible Python pipeline. Given a target gene, the pipeline proceeds through the stages described below. It is built on Biopython (**NCBI** access and sequence handling), an external MAFFT executable (alignment), `primer3-py` (thermodynamics), `seqfold` (secondary structure) and `pandas` (tabular output); the three-dimensional validation is run separately with Boltz-2 on a **Graphics Processing Unit (GPU)**. Every design threshold is resolved through a layered scheme that can be overridden per gene, per species and per organism type (Section 3.8).

## 3.1 Sequence retrieval from NCBI

For each target, sequences are retrieved from the **NCBI** nucleotide database through the Entrez interface. An `esearch` query returns up to a configurable number of identifiers (set to $100$ in this work; justified in Section 3.12), which are then fetched with `efetch` in FASTA format, in batches of $200$ identifiers, with short pauses between requests to respect the **NCBI** rate limit of three requests per second without an API key. Each target defines a primary query combining the gene name, the organism and a sequence-length range; if it returns fewer than two records, the pipeline falls back in turn to one or two broader alternative queries and, failing those, to a fixed RefSeq accession. Records outside a coarse per-gene length range are discarded at this stage. For genes with a single usable record (allowed only for *rmpM*), a single-sequence mode is used.

## 3.2 Length clustering

Public records mix full-length genes with partial fragments and oversized entries, which together produce gap-ridden alignments. An automatic length-clustering step therefore retains only the dominant length cluster: the sequence length whose $\pm 25\%$ band contains the most records is taken as the centre, and only records within that band are kept. The selection is data-driven — no per-gene minimum or maximum length is tuned by hand — so the method transfers to new genes without adjustment. It is applied only when at least four records are available, and a safeguard prevents the set from being reduced below two sequences.

## 3.3 Multiple sequence alignment

The retained sequences are aligned with MAFFT in its automatic mode (`--auto`). When only one sequence is available the alignment step is bypassed and that sequence is used directly.

## 3.4 Conservation scoring (PPI)

Conservation is scored per alignment column using the **Percentage of Pairwise Identity (PPI)**, adopted from the *ViruScope* tool (Lima, 2023). For a column, the **PPI** is the average identity over all pairs of non-gap bases, computed from the base counts; pairs of ambiguous **IUPAC** bases contribute a partial weight (for example a purine code $R$ scores $0.5$ against $A$ or $G$, and $N$ scores $0.25$ against any base). Columns in which fewer than $30\%$ of the sequences have a non-gap base are assigned zero, and only columns with at least two non-gap bases are counted. Because the per-column **PPI** is independent of the window, it is pre-computed once per alignment, which is essential for hundreds of sequences. The conservation of a candidate window is the mean of the **PPI** of its valid columns. In contrast to primer design, the identity of the terminal $3'$ nucleotides (scored separately as PPI3 in the original tool) is deliberately excluded, because capture probes are not extended by a polymerase and so the fidelity of their $3'$ end is irrelevant.

## 3.5 Candidate generation

Candidate probes are generated by sliding a window over the alignment. For each start position, taken every $3$ columns, windows of every length from $18$ to $28$ nucleotides are formed. A window is discarded if any of its columns has a gap fraction above the allowed maximum ($20\%$ by default), or if its mean conservation is below the target's conservation threshold. The sequence of a surviving window is the per-column consensus (the most frequent non-gap base).

## 3.6 Thermodynamic screening

Each candidate is scored with `primer3-py` under fixed conditions: a monovalent cation concentration of $50$ mM, no divalent cations or dNTPs, an oligonucleotide concentration of $250$ nM and a reference temperature of $37\,°$C. Four quantities are computed: the melting temperature $T_m$ (**Nearest-Neighbour (NN)** model (SantaLucia Jr. and Hicks, 2004)), the GC content (the fraction of G and C bases), the hairpin free energy and the homodimer (self-dimer) free energy, the last two reported in kcal/mol and only when a structure is found. A candidate passes the basic screen when its $T_m$ and GC content fall within the resolved windows and its hairpin and homodimer free energies are no more negative than their respective limits (Table 2). The $T_m$ window may optionally be derived from the hybridisation assay temperature $T$ as $[T + 15,\ T + 35]\,°$C.

| **Parameter** | **Default value** | **Basis** |
|---|---|---|
| Probe length | 18–28 nt | Wetmur (1991) |
| Melting temperature $T_m$ | 53–72 °C | SantaLucia Jr. and Hicks (2004); assay-derivable |
| GC content | 40–60% | IDT OligoAnalyzer (Integrated DNA Technologies) |
| Hairpin $\Delta G$ | $\geq -2.0$ kcal/mol | Integrated DNA Technologies; primer3 (Untergasser et al., 2012) |
| Homodimer $\Delta G$ | $\geq -5.0$ kcal/mol | IDT OligoAnalyzer (Integrated DNA Technologies) |
| Conservation (**PPI**) | $\geq 0.85$ | pan-strain detection (Lima, 2023) |
| Alignment gap fraction | $\leq 20\%$ | alignment quality |
| Secondary structure $\Delta G_{\text{MFE}}$ | $\geq -2.6$ kcal/mol | 5th percentile of the data |

Table 2: Global default thresholds. Values are overridden per organism type, species or gene where justified (Section 3.8).

## 3.7 Secondary structure and the No-fold score

The single-strand secondary structure of each candidate that passed the basic screen is computed with `seqfold`, which returns the **Minimum Free Energy (MFE)** at $37\,°$C and the fraction of paired bases. A candidate passes the secondary-structure screen when its **MFE** is no more negative than $-2.6$ kcal/mol; this value is the fifth percentile of the observed **MFE** distribution (Figure 1), and fixing it keeps the criterion reproducible across runs rather than always cutting a fixed fraction of the data.

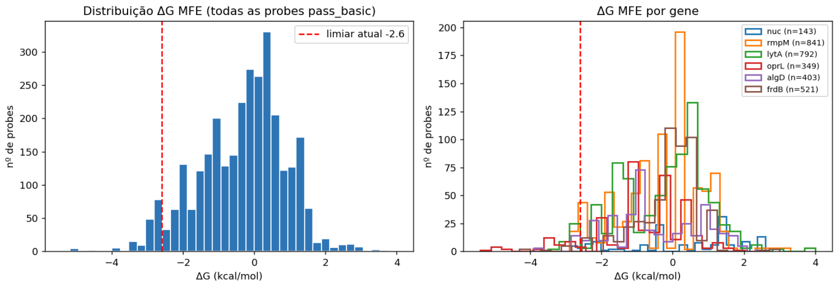


Figure 1: Distribution of the `seqfold` secondary-structure free energy ($\Delta G_{\text{MFE}}$) over the candidate probes, from which the $-2.6$ kcal/mol threshold (fifth percentile) is taken.

In addition to this pass/fail screen, the three folding energies (hairpin, homodimer and **MFE**) are combined into a continuous *No-fold* score in the range $0$–$100$, following *ViruScope* (Lima, 2023). Each energy is mapped through a logistic penalty,

$$\text{No-fold} = \left(1 - \frac{1}{1 + \exp(x)}\right) \times 100, \qquad x = \frac{\Delta G_{\text{cal}} - \Delta G_0/2}{s\,RT},$$

with $\Delta G_0 = -7000$ cal/mol, slope $s = 1$, $R = 1.987$ cal mol$^{-1}$K$^{-1}$ and $T = 310$ K; the per-probe No-fold score is the mean of the scores of the energies that are available. A high No-fold score denotes a probe that is largely free of stable self-structure. This continuous score is used, together with conservation, to rank the candidates rather than merely to filter them.

## 3.8 Organism-aware parameters

Because the intended application spans pathogens with very different genomes, a single global set of thresholds is inadequate: criteria appropriate for a balanced bacterial genome would unfairly reject an AT-rich or a GC-rich target. All thresholds are therefore resolved from the most specific layer to the most general,

$$\text{gene} \;\rightarrow\; \text{species} \;\rightarrow\; \text{organism type} \;\rightarrow\; \text{global default},$$

where the first layer that defines a parameter wins. Five organism types are distinguished, differing mainly in genome GC content and in variability (Table 3); species-level profiles then fine-tune the values by the GC content of each genome (for example *P. aeruginosa* at $\sim$67% GC (Stover et al., 2000), *A. fumigatus* at $\sim$50% GC (Nierman et al., 2005) and the AT-rich *P. falciparum* at $\sim$19% GC (Gardner et al., 2002)).

| Organism type | GC window | Conservation | Max length |
|---|---|---|---|
| Bacteria | 40–60% | 0.85 | 28 |
| Virus | 35–68% | 0.70 | 30 |
| Fungus | 40–60% | 0.80 | 28 |
| Protozoan | 15–40% | 0.80 | 30 |
| Host (human) | 35–60% | 0.85 | 28 |

Table 3: Per-type parameter profiles. Viruses use a wider GC window and a more permissive conservation threshold; the protozoan profile is AT-shifted.

When the pipeline encounters an organism for which no profile exists and is run interactively, it prompts the user for the organism type and for each threshold, proposing a suggested value that can be accepted or overridden; the resulting profile is stored and reused in later runs. When run non-interactively, it applies the suggested values without prompting.

## 3.9 Quality-based selection

Because three-dimensional prediction is comparatively expensive, only the best candidates per gene are carried forward. Rather than ranking by melting temperature — which the thermodynamic screen has already bounded — the surviving candidates are ranked by a *quality* score that averages the two signals most indicative of sequence quality:

$$\text{quality} = \tfrac{1}{2}\left(\textbf{PPI} + \tfrac{\text{No-fold}}{100}\right).$$

The top-ranked candidates of each gene are exported for 3D validation, and the same ranking is used to export larger curated sets (for example the top $25$ per gene) together with their sequence-level features.

## 3.10 Three-dimensional validation

The selected probes are written one per file in the native Boltz-2 input format and submitted to Boltz-2 (Wohlwend et al., 2024), run on a **GPU**, which predicts the three-dimensional structure of each single-stranded sequence together with confidence estimates: an overall confidence, the **predicted Local Distance Difference Test (pLDDT)** and the **Predicted Aligned Error (PAE)** matrix, in the style of AlphaFold 3 (Abramson et al., 2024). As these confidence metrics are calibrated primarily for proteins, the overall confidence and the **pLDDT** are used here as relative, independent indicators of the structural soundness of the short probes. The predictions are merged with the sequence metadata to produce a final ranked shortlist combining sequence quality (conservation and No-fold) with the 3D confidence.

## 3.11 Integration of a reference probe panel

An existing experimental probe panel was integrated into the same workflow for comparison. Each reference probe, supplied as a (`job_name`, `sequence`, `species`) record, is cleaned (uracil converted to thymine, non-ACGT characters removed) and scored with exactly the same metrics as the designed probes, but by the thresholds of its *own* species and type. Conservation is left empty for these probes, because it is only definable for probes derived from an alignment. The reference probes are stored alongside the designed ones, with a source label, so that the two sets appear side by side with identical columns.

## 3.12 Diversity and rarefaction analysis

To characterise each target and to justify the sampling depth, an alignment-free analysis based on $4$-mer composition is performed on the retrieved sequences. Each sequence is represented by its vector of $4$-mer frequencies, and pairwise cosine distances (on the frequency vectors) and Jaccard distances (on $4$-mer presence/absence) are computed, together with the fraction of unique sequences, which detects exact duplicates. A rarefaction analysis subsamples $N$ sequences per gene and measures the mean $4$-mer diversity and the richness (number of distinct $4$-mers), to determine the value of $N$ beyond which little new information is gained.

## 3.13 Implementation and reproducibility

All intermediate and final results are written as CSV files. Because **NCBI** does not always return an identical set of records, per-gene counts may vary by a few percent between runs; for strict reproducibility the accession identifiers can be frozen.

# Chapter 4

# Results and Discussion

This chapter reports the results obtained by applying the pipeline to the six-gene panel, with $100$ sequences requested per gene (and a per-gene override of $200$ for the data-limited *algD*). It follows the flow of the method: sequence retrieval and curation, the screening funnel, three-dimensional validation, the diversity and rarefaction analyses, the effect of the organism-aware parameters, the comparison with the reference panel, and the curated dataset of prioritised candidates.

## 4.1 Retrieval and length curation

Table 4 shows, for each gene, the number of sequences returned by **NCBI** and the number retained after length clustering. The clustering step removed partial and oversized records that would otherwise degrade the alignment; its effect is largest for the genes with the most heterogeneous records. Two genes, *oprL* and *algD*, are intrinsically data-limited in **NCBI** (only 24 and 12 sequences survive respectively), which, as shown below, is reflected in their behaviour through the pipeline.

| Gene | Retrieved | After clustering | Alignment (cols) |
|---|---|---|---|
| *nuc* | 99 | 77 | 517 |
| *rmpM* | 100 | 76 | 1311 |
| *lytA* | 95 | 87 | 2594 |
| *oprL* | 27 | 24 | 770 |
| *algD* | 19 | 12 | 1020 |
| *frdB* | 82 | 82 | 489 |

Table 4: Sequences retrieved and retained per gene, and the resulting alignment width.

## 4.2 Screening funnel

Across the six genes, the pipeline generated $8455$ candidate windows and filtered them down to $2943$ that passed both the thermodynamic screen and the secondary-structure screen (Table 5). The melting-temperature and GC filters remove the largest fractions, as expected, since they encode the core hybridisation requirements; the homodimer filter is particularly severe for the GC-rich *P. aeruginosa* genes (*oprL*, *algD*), whose sequences are more prone to self-complementarity. The secondary-structure screen removes only a small additional fraction, confirming that most structurally problematic candidates are already eliminated by the thermodynamic filters.

| Gene | Windows | $T_m$ | GC | Hairpin | Basic | Seqfold |
|---|---|---|---|---|---|---|
| *nuc* | 667 | 326 | 154 | 143 | 143 | 138 |
| *rmpM* | 1867 | 1621 | 1366 | 1251 | 841 | 823 |
| *lytA* | 1833 | 1274 | 966 | 942 | 820 | 771 |
| *oprL* | 1360 | 1273 | 990 | 800 | 349 | 309 |
| *algD* | 1012 | 942 | 763 | 662 | 403 | 389 |
| *frdB* | 1716 | 932 | 752 | 733 | 522 | 513 |
| **Total** | **8455** | **6368** | **4991** | **4531** | **3078** | **2943** |

Table 5: Screening funnel: candidate windows surviving each successive filter. "Basic" is the number passing all four thermodynamic criteria; "Seqfold" additionally passes the secondary-structure screen.

## 4.3 Three-dimensional validation

From the candidates that passed screening, the five highest-quality per gene ($30$ probes in total) were selected by the quality score and validated with Boltz-2. Of these, $4$ were classified as good, $11$ as moderate and $15$ as low by the overall confidence. The best candidates, strong *both* in sequence quality and in 3D, are listed in Table 6; they come from *algD* and *frdB*. The selected probes are strongly conserved (**PPI** up to $1.00$) and essentially free of self-structure (No-fold above $96$), which coincides with the highest structure-prediction confidence and **pLDDT** among the $30$ validated probes. The agreement between the sequence-level signals and the independent 3D prediction provides an internal consistency check on the ranking.

| Probe | Gene | PPI | No-fold | Confidence | pLDDT |
|---|---|---|---|---|---|
| p5972 | *algD* | 0.93 | 98.4 | 0.731 | 0.901 |
| p7716 | *frdB* | 0.99 | 99.6 | 0.723 | 0.822 |
| p5938 | *algD* | 0.90 | 96.2 | 0.721 | 0.887 |
| p8448 | *frdB* | 1.00 | 98.2 | 0.716 | 0.845 |

Table 6: Top probe candidates, strong in both sequence quality and 3D confidence.

## 4.4 Genetic diversity of the targets

The alignment-free diversity analysis (Table 7) explains much of the behaviour observed above. *lytA* is by far the most diverse gene (mean cosine distance $0.187$ and $93\%$ unique sequences), consistent with its known allelic diversity in *S. pneumoniae* (Obregón et al., 2002); this justifies the relaxed conservation threshold applied to it. At the opposite extreme, *frdB* is the most conserved and most redundant gene (mean cosine distance $0.025$ and only $35\%$ unique sequences, i.e. many exact duplicates in **NCBI**), which is why it yields probes of very high conservation. *algD* has $100\%$ unique sequences but very few of them, which accounts for its instability at low sampling and motivates the automatic length-clustering and the per-gene allowance of more sequences. The cosine and Jaccard distances are complementary — the former weighs $4$-mer frequencies, the latter only their presence or absence — and they concur in identifying *frdB* as the most conserved and redundant gene, giving an alignment-independent reading of diversity.

| Gene | $n$ | Cosine (mean) | Cosine (max) | Jaccard (mean) | Unique (%) |
|---|---|---|---|---|---|
| *nuc* | 77 | 0.059 | 0.425 | 0.163 | 61.0 |
| *rmpM* | 76 | 0.079 | 0.247 | 0.069 | 50.0 |
| *lytA* | 87 | **0.187** | 0.440 | 0.125 | 93.1 |
| *oprL* | 24 | 0.050 | 0.358 | 0.096 | 91.7 |
| *algD* | 12 | 0.097 | 0.332 | 0.117 | **100.0** |
| *frdB* | 82 | **0.025** | 0.052 | 0.061 | **35.4** |

Table 7: Alignment-free $4$-mer diversity per gene. Bold marks the most diverse (*lytA*) and the most conserved/redundant (*frdB*) genes.

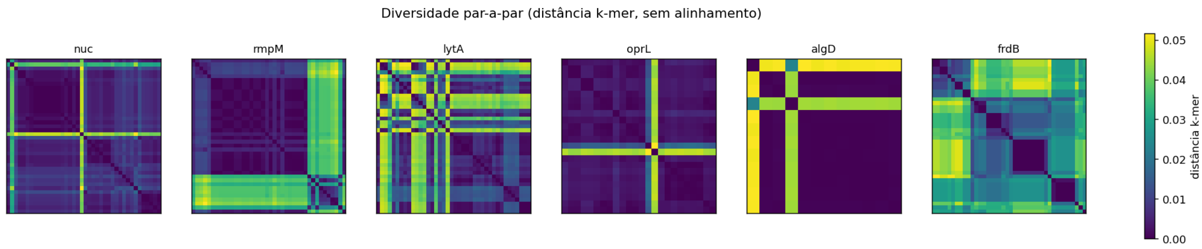


Figure 2: Pairwise $4$-mer distance heatmaps per gene (alignment-free). Each cell is the $4$-mer distance between two retrieved sequences of the gene.

## 4.5 Choice of sampling depth

The rarefaction analysis showed that both the $4$-mer diversity and the $4$-mer richness saturate at around $N \approx 50$ sequences for the most demanding gene (*nuc*), and much earlier for the others. Requesting up to $100$ sequences per gene therefore captures essentially all of the available diversity with a comfortable margin, while additional sequences add little new information. The true limiting factor is not this cap but the small number of records available in **NCBI** for the data-limited genes.

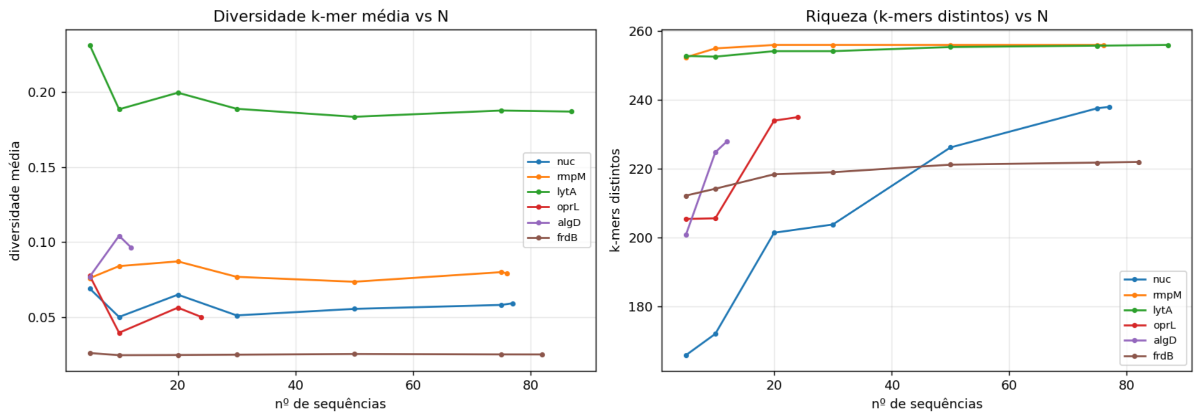


Figure 3: Rarefaction of the mean $4$-mer diversity (left) and of the richness, i.e. the number of distinct $4$-mers (right), per gene, as a function of the number of subsampled sequences.

## 4.6 Effect of the organism-aware parameters

The value of resolving thresholds per organism type and species is clearest when scoring the reference panel, which spans bacteria, viruses, a fungus, a protozoan and a human control. When the $74$ reference probes were scored with a single global (bacterial) criterion, $29$ passed the basic screen; when instead each was scored by the criteria of its own species and type, $39$ passed. The additional probes are precisely those whose genome composition differs from the balanced bacterial case — for example the AT-rich *Plasmodium* probe, which the global GC window would reject but which is appropriate under the protozoan profile. This confirms that organism-aware thresholds avoid unfair rejections without loosening the criteria for the balanced targets.

## 4.7 Comparison with the reference panel

Scored with identical metrics, the designed probes and the reference panel can be compared directly. Among the probes that pass the basic screen, the designed probes have a higher mean melting temperature ($59.3$ versus $56.9\,^{\circ}$C) and a very similar mean GC content ($0.52$ versus $0.53$), while the reference probes have a marginally higher mean No-fold score ($86.0$ versus $82.9$). Because both sets carry a source label and share every column, they can also be submitted together to Boltz-2, enabling a like-for-like comparison in three dimensions as well as in sequence space.

## 4.8 Curated dataset of prioritised candidates

Finally, the same quality ranking was used to export a larger curated dataset: the top $25$ probes per gene ($150$ probes in total), each with its sequence-level features ($T_m$, GC, conservation, No-fold and folding energies). This dataset extends the set of $30$ probes validated in 3D to a broader, biophysically annotated selection of the highest-quality candidates per gene.

# Chapter 5

# Conclusions and Future Work

## 5.1 Conclusions

This dissertation set out to design, *in silico*, high-quality **ssDNA** capture probes for **GFET** biosensors targeting a panel of clinically relevant bacterial pathogens. The main outcome is a single, reproducible computational pipeline that turns a target gene into a small set of ranked, biophysically validated probe candidates, with every design decision grounded in the literature.

The pipeline integrates, in one automated workflow, the stages that are usually performed separately or by hand: retrieval and curation of sequences from **NCBI**, automatic length clustering, multiple sequence alignment and **PPI**-based conservation scoring, nearest-neighbour thermodynamic screening, secondary-structure evaluation with a continuous No-fold score, and a final three-dimensional validation with Boltz-2. Two design choices give the method its generality and rigour. First, thresholds are resolved through a layered, *organism-aware* scheme (gene $\rightarrow$ species $\rightarrow$ type $\rightarrow$ global), so that pathogens with very different genome composition are judged by fair criteria; this was shown to recover candidates — such as AT-rich probes — that a single global rule would wrongly discard. Second, candidates are prioritised by a transparent quality score combining conservation and the No-fold score, which concentrates the expensive 3D validation on the most promising sequences.

Applied to the six-gene panel, the pipeline reduced $8455$ candidate windows to $2943$ that passed screening and identified a shortlist of top candidates whose high sequence quality was independently corroborated by the highest 3D confidence and **pLDDT**. Alignment-free diversity and rarefaction analyses characterised each target and justified the sampling depth in a principled, non-arbitrary way. An existing experimental probe panel was folded into the same workflow and scored with identical metrics, enabling a direct comparison, and a curated $150$-probe dataset of the highest-quality candidates per gene was produced.

Overall, the work delivers a solid, extensible foundation for the computational stage of **GFET** biosensor development: it produces prioritised, structurally sound probe candidates and generalises to new genes and organisms with minimal configuration.

## 5.2 Prospects for future work

Several directions follow naturally from this work:

- **Experimental validation.** The ultimate test of the designed probes is their performance on fabricated **GFET** devices; the ranked shortlist provides the immediate candidates for functionalisation and characterisation.
- **Specificity screening.** Adding an automated specificity check (for example a BLAST step against non-target genomes and the host) would complement the current conservation-driven, sensitivity-oriented selection.
- **Comparison of 3D methods.** The Boltz-2 predictions can be benchmarked against an independent molecular-docking pipeline, to assess the agreement of the two structural views of the probe–target interaction.
- **Data-driven modelling.** The curated feature dataset opens the way to a machine- learning model that predicts probe quality directly, potentially accelerating or refining the ranking.
- **Broader panels.** As the framework already supports viral, fungal, protozoan and host profiles, extending the panel to further genes and non-bacterial pathogens is largely a matter of configuration.

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